

A1 DIAGNOSIS

See the Future Clearly

The First AI-Powered Blood-Based Risk Assessment Test
For Age-Related Macular Degeneration

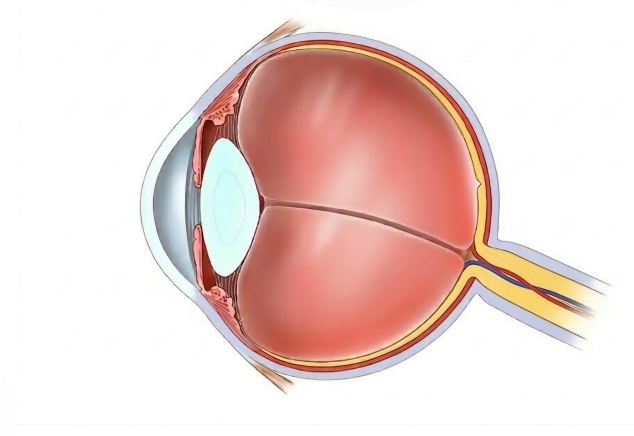
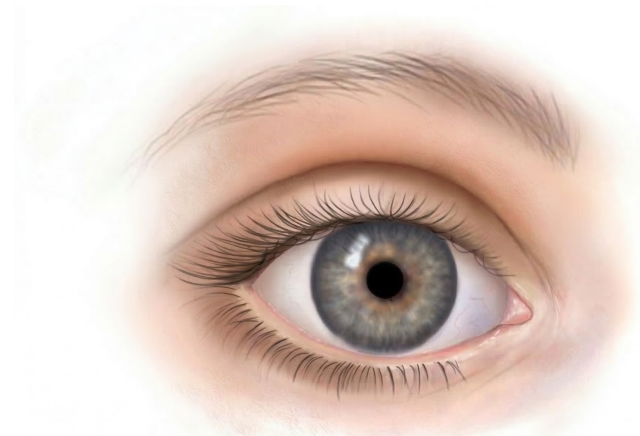
PRE-SEED ROUND
\$1.5M

Mustafa Ozgul, MD
CEO & Founder
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Leading Cause of Vision Loss

in United States



United States Impact

20M

Americans Affected by AMD



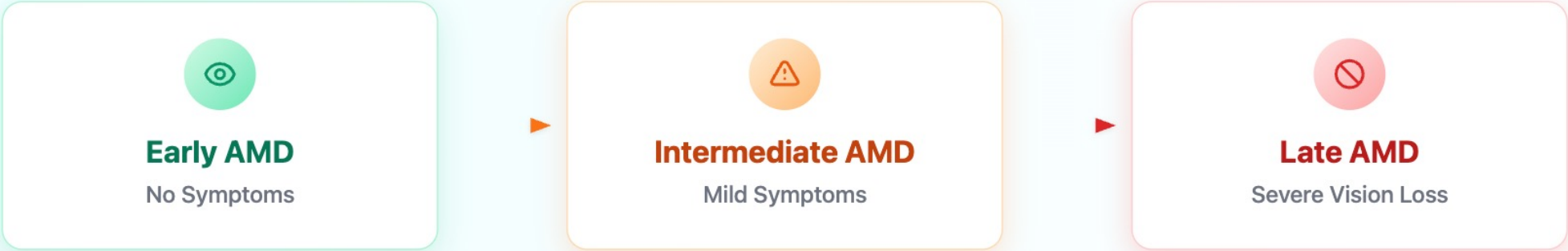
Global Impact

200M

People Affected Worldwide



Disease Progression



● Normal Vision

VS



● AMD Vision Loss

New Therapies Coming for Early AMD

DRUG	SPONSOR	PHASE	ROUTE	TARGET
Iptacopan NCT05230537	Novartis	Phase 2	Oral	Early / Intermediate
Atorvastatin NCT04735263	Mass Eye & Ear	Phase 2	Oral	Intermediate
Aspirin + Clopidogrel + NAC NCT06165068	Navamindrathiraj University	Phase 3	Oral	Intermediate

How It Works?



Eye Doctor Visit

1



Sample Collection

2



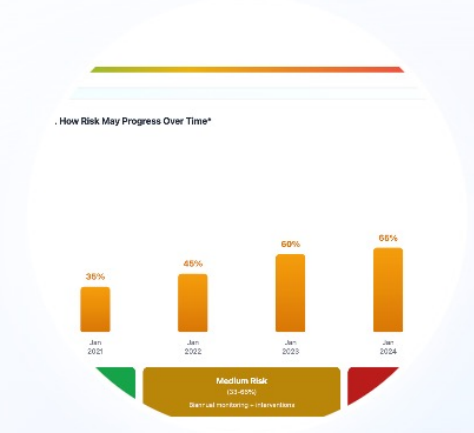
**Mass Spectrometry
with Attomole
Sensitivity**

3



**Machine Learning
Analysis**

4



Risk Assessment

5

Proprietary AI/ML Platform

>95%

ML Accuracy

ROC AUC 0.987

398

Biomarkers Analyzed

10K+

Peptides Detected



6-Year Research Moat

UC Irvine Partnership

Gavin Herbert Eye Institute — specialized AMD biomarker research program

15+ Publications

Peer-reviewed research validating native peptide analysis methodology

First-to-Market

No one else doing native peptide analysis for AMD Early Detection

EXCLUSIVE



FIRST-EVER study identifying native peptide signatures in wet AMD plasma

Fundus Photography

Standard retinal scan at eye exam

~8%

Drusen Detection in Early AMD
Color fundus photography (CFP)

WHAT IT DETECTS

✗ Visible drusen and pigment changes only

WHEN IT DETECTS

✗ After structural damage is already visible

LIMITATION

✗ Cannot detect RPE dysfunction or cell death before drusen form

COST

\$25 – \$50 per scan

AI-ENHANCED (BEST CASE)

88% sensitivity — but **only for intermediate + advanced AMD**. Early AMD classified as negative.

Dong et al. EClinicalMedicine 2021 (1.2M+ images, 19 studies). Kiruthika & Malathi, Photodiagnosis Photodyn Ther 2025.

Optical Coherence Tomography

OCT — gold standard retinal imaging

14.9% – 68.1%

Sensitivity for Early AMD
Cirrus (14.9%) → Dense Spectralis (68.1%)

WHAT IT DETECTS

△ Drusen volume, RPE elevation — structural changes only

WHEN IT DETECTS

△ After drusen deposits have formed

LIMITATION

✗ Sees RPE layer but cannot distinguish healthy vs. dying cells within it

COST

\$40 – \$100 add-on to eye exam

KEY FINDING

Best OCT (dense Spectralis 241 B-scans) still **misses 32% of early AMD**. Standard clinical Cirrus **misses 85%**.

Garzone et al. "MACUSTAR study report." Scientific Reports 12, 21911 (2022). 380 eyes, 200 individuals, 20 sites, 7 countries.

AI Diagnosis Blood Test

AI-powered native peptide risk assessment

>95%

ML Accuracy in Proof of Concept
ROC AUC 0.987 | 398 biomarkers

WHAT IT DETECTS

✓ RPE dysfunction and cell death products — native peptides in blood

WHEN IT DETECTS

✓ **Before** structural damage is visible on any imaging

ADVANTAGE

✓ Measures biology, not structure — detects what OCT and funduscopy fundamentally cannot

COST

\$250 per test — standard blood draw

INFRASTRUCTURE

✓ No imaging device needed. Simple blood draw at routine eye visit.

AI Diagnosis proof-of-concept. 12 samples, 398 biomarkers, 97.8% ML accuracy. 240-sample validation planned.

OCT sees the RPE layer perfectly. What it cannot see is which cells are dying. AI detects what those cells release into the blood.

Market Opportunity



TAM
\$42.3B

169M Americans 35+ x \$250/test



SAM
~\$6.0B

23.4M adults 45+ with eye care & clinical ordering



SOM
~\$100M
Year 5 Target

\$1M (Y1) → \$100M (Y5) · 100x growth · bottom-up from clinic volume

▼ SAM Methodology

Filter 1: Age 45+ Eligible for AMD Risk Assessment

~127M

Filter 2: Annual Eye Care Visit (~56% of Patients 45+ Years Old)

~71M

Filter 3: Clinical Ordering Rate (~33%)

Published clinical risk factors for AMD

~23.4M

$\text{SAM} = 23.4\text{M} \times \$250 = \text{~\$6.0 Billion}$

Sources: U.S. Census Bureau, CDC VEHSS, Ehrlich et al. JAMA Ophthalmology 2019, Saydah et al. 2020

Business Model & Unit Economics

How We Make Money

REVENUE STARTS Q1 2027

B2B MODEL



Routine Eye
Doctor Visit



A1 CLIA Lab
Analyzes & Reports



A1 invoices clinic directly
\$250 / test

YEAR 1-2
PLA code application
Medicare via MoIDX/ADLT

POST-FDA CLEARANCE
Private insurance
coverage

Unit Economics per Test

\$250

Revenue

Per Test

-\$60

COGS

\$190

Contribution

76% Margin

-\$50

Fixed Costs

\$140

Gross Profit

56% at Scale

At 500+ tests/month

GROSS MARGIN BENCHMARKS AT SCALE

Exact Sciences

69%

Guardant Health

62%

A1 Diagnosis

56%

→ 65-70% target

Go-To-Market Strategy

B2B through optometry & ophthalmology practices

1



Seed Clinics

15 partner clinics in Southern California



2



Regional Growth

Conference outreach
KOL partnerships



3



Reimbursement Unlock

PLA code
Medicare drives adoption



4



National Scale

1,200 clinics post-FDA clearance

B2B Only



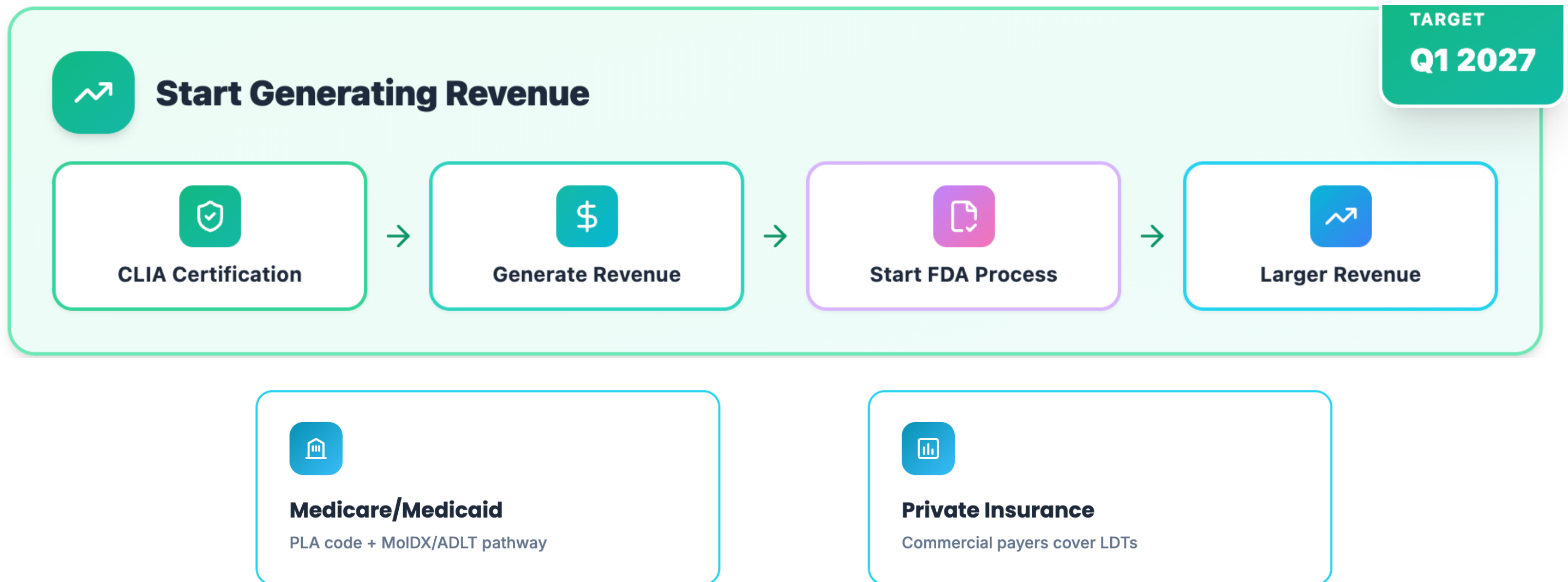
\$250 / Test Invoiced to Clinics



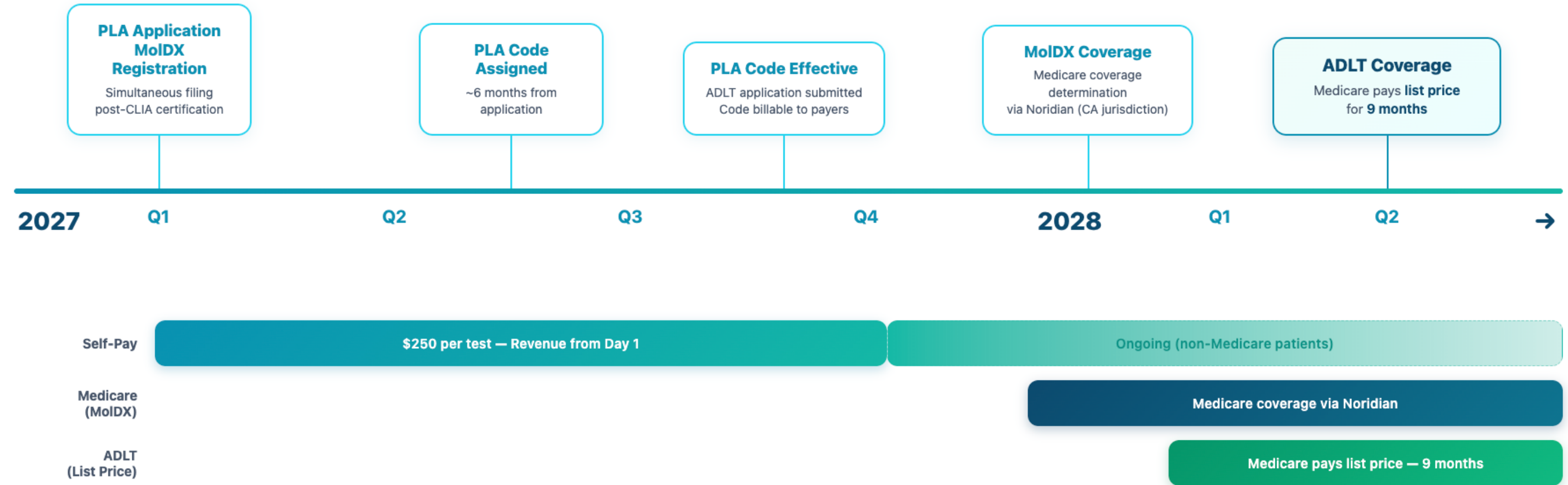
Routine Eye Doctor Visit

De-risked Revenue Path

Faster time-to-market than FDA approval



Reimbursement Timeline



Competitive Landscape

First blood biomarker for AMD risk assessment

Primarily DR — one AMD 510(k) pending, no blood-based AMD test in US

\$200M+ invested in AI imaging

0 FDA-cleared blood-based AMD risk assessment in US



AI Retinal Imaging

DR products only — no AMD capability

Digital Diagnostics · \$130M+Eyenuk · \$43M+AEYE Health · \$28M+

iHealthScreen · 510(k) pending

AMD PRODUCT (US)



✗ None FDA-cleared

iHealthScreen iPredict-AMD: 510(k) submitted May 2023 — pending (86.9% sens, 94.1% spec)

WHEN IT DETECTS



Too late — after visible retinal damage

WHAT IT MEASURES



Visible structural damage in camera
Drusen, lesions

INFRASTRUCTURE & ACCESS



Retinal imaging device required in doctor office

iHealthScreen requires iCare DRSPlus fundus camera + trained operator



Genetic Testing

Inherited DNA — what you're born with

Macula Risk PGx · Arctic MedicalAMDiGuard · Visible Genomics

**AAO 2025**
Not recommended

AMD PRODUCT (US)



Risk score using DNA (CLIA LDT)

WHEN IT DETECTS



Lifetime risk — born with it

WHAT IT MEASURES



Static inherited risk — 15+ SNPs

INFRASTRUCTURE & ACCESS



Cheek swab — mail-in kit, \$229–\$400



AI Powered Blood Test

Risk Assessment — before vision loss

A1 Diagnosis · Pre-Seed

FIRST & ONLY

AMD PRODUCT (US)



✓ First blood-based AMD biomarker

WHEN IT DETECTS



Before vision loss

WHAT IT MEASURES



Active RPE cell death — not static DNA, not visible damage

INFRASTRUCTURE & ACCESS



Standard blood draw — A1 Diagnosis CLIA lab

Risks & Mitigation

● RISK

Early-stage proof of concept (n=12)

↑ OUR STRENGTH

Like spotting a red ball in a bag of white ones — the biomarker signal is unmistakable, even in small samples

✓ MITIGATION

240-sample validation designed & powered. Funded by this raise.

● RISK

Provisional patents not yet filed

↑ OUR STRENGTH

6-year native peptide research moat — no competing AMD blood test exists

✓ MITIGATION

Patent filing is a funded milestone. Trade secret protections active.

● RISK

Eye care clinics lack trained phlebotomists

↑ OUR STRENGTH

\$250 test, simple workflow, fits routine eye visit

✓ MITIGATION

DARPA-funded Tasso device enables painless collection — no phlebotomist needed. PLA code + Medicare pathway.

Growth Projections

Revenue Growth



	CLIA LDT Launch	CLIA LDT Scale	De Novo Sub.	FDA + Insurance	National Scale
Partner Clinics	15→30	75	150→300	600	1,200
Tests/Mo	450	1,125	6,600	15,000	33,000
Annual Tests	4,050	13,500	55,800	180,000	400,000

Investment Allocation for 18 Months



Regulatory & Certification

ResearchDx CLIA + CAP	\$140,000
CLIA/CA Application Fees	\$7,143
CAP Fees	\$11,250
D&O Insurance (\$250/mo × 18)	\$4,500
Legal & IP	\$25,000
Contingency	\$16,000
Subtotal	\$203,893



Personnel

Founder/CEO (\$100K/yr × 1.5)	\$150,000
Lab Director (consultant, \$5K/mo × 18)	\$90,000
CLS Tech (part-time 49%, \$3.2K/mo × 18)	\$57,600
Subtotal	\$297,600



Equipment & Lab Space

Lab Space (\$4K/mo × 18)	\$72,000
timsTOF HT Lease (\$28K/mo × 18)	\$504,000
Ancillary Equipment	\$50,000
Subtotal	\$626,000



Validation, IT & Operations

Clinical Samples (240)	\$69,600
QC Materials (\$5K/mo × 18)	\$90,000
LIS + Interface (\$2.5K/mo × 18)	\$45,000
Compute/Storage	\$29,000
Business Ops (\$625/mo × 18)	\$11,250
Contingency	\$127,657
Subtotal	\$372,507

Pre-seed / SAFE

18-Month Total Investment

\$1,500,000

Our Founding Team



Dr. Mustafa Oztgul

Founder & CEO

 [linkedin.com/in/drozgul](https://www.linkedin.com/in/drozgul)

Medical Doctor and scientist who analyzed potential AMD biomarkers at UC Irvine in Gavin Herbert Eye Institute for nearly 6 years using state-of-art techniques.

Currently developing groundbreaking machine learning algorithms analyzing up to 10,000 natural molecules for early AMD Risk Analysis Platform.

- 💡 **UC Irvine Gavin Herbert Eye Institute** - 6 years of AMD research
- 💡 **Discovery Center for Eye Research** Grant Recipient
- 💡 **ARVO Conference** - Multiple publications and presentations
- 💡 **10,000 molecules** - ML/DL algorithms development
- 💡 **US-funded Research Projects** - Principal Investigator (PI)



Dr. Murat Baday

Founder

Stanford Scientist & Synapses Ventures Founder

 [linkedin.com/in/muratbaday](https://www.linkedin.com/in/muratbaday)

Stanford researcher and Synapses Ventures founder with proven medical device commercialization success.

Led Smartlens Inc., developer of the world's first electronics-free contact lens for glaucoma monitoring, to \$6.1M Series A funding (July 2023).

- 💡 **Smartlens Inc.** - \$6.1M Series A (July 2023)
- 💡 **Stanford Researcher** - Computational Biology & Biophysics PhD
- 💡 **40+ Patents** - Innovative medical device solutions
- 💡 **1,259 Research Citations** - Proven scientific impact
- 💡 **Synapses Ventures** - Founder & Innovation Leader

Exit Strategy

ACQUISITIONS

YEAR	COMPANY	DESCRIPTION	CLIA LDT	ACQUIRED BY	REVENUE	AMOUNT	KEY METRIC
2014	Ariosa Diagnostics	Risk assessment test for prenatal conditions	✓	Roche	Pre-revenue	\$625M	Early-stage, strategic
2016	Sequenom	Risk assessment test for prenatal conditions	✓	LabCorp	\$128M	\$371M	2.9× revenue
2019	Genomic Health	Risk assessment test for cancer recurrence	✓	Exact Sciences	\$452M	\$2.8B	4.9× revenue

IPOS

YEAR	COMPANY	DESCRIPTION	CLIA LDT	VEHICLE	REVENUE	MARKET CAP AT IPO	CURRENT
2014	CareDx	Risk assessment test for transplant rejection	✓	Nasdaq	\$23M	\$115M	~\$870M
2015	Natera	Risk assessment test for prenatal + transplant conditions	✓	Nasdaq	\$159M	\$1.1B	~\$26.6B

Summary & Thank You

Let's Build the Future of Eye Health Together

✓ **First blood-based AMD risk assessment platform**

New category — no competitor exists

✓ **> 95% accuracy in proof-of-concept**

15+ peer-reviewed AMD publications

✓ **\$42.3B TAM — 169M Americans at risk**

Growing as population ages

✓ **Revenue Q1 2027 — CLIA-first model**

No FDA approval needed to launch

✓ **\$1M → \$100M in 5 years**

Bottom-up: 15 to 1,200 partner clinics

PRE-SEED ROUND

\$1.5M SAFE

- \$8M Pre-Money cap
- 18-month runway through CLIA cert & first revenue

Dr. Mustafa Ozgul

Founder & CEO

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